# Lab 4.4 Researcher and Analyst

IES | 15 minutes | LO5

All records are fictional training data. They describe no actual facility, person or organisation. Values are teaching assumptions, not approved design criteria. Do not use outputs to operate, maintain or approve real equipment.

## Inputs and preparation

Open the mock-data folder: research-question.txt, process-data.csv, defects.csv, claim-register.csv. Use prompts.md for copy and paste. Save your reviewed work under outputs. Earlier lab outputs are optional; this folder provides standalone inputs.

## Procedure

1. In Microsoft 365 Copilot, select Researcher from Agents if available. Paste research-question.txt. Require sources supporting each comparison claim and record scope limitations.

2. Open two cited source links. Check publisher, title, date or revision and whether the source actually supports the associated statement. Catalogue pages do not provide full standard clauses.

3. Select Analyst and attach process-data.csv and defects.csv. Use the prompt below. Require baseline and monitoring separation and visible calculations.

# Lab 4.4 continued

4. Compare Analyst results with the formula method in this folder expected-output.md. Challenge any claim that a signal proves a particular root cause.

5. Save a research comparison, two verified citations and the analysis with a correction log. If either agent is unavailable, run a paired demonstration or manual source analysis and record the missing live capability.

## Check before moving on

At least two citations are opened and checked.

Sample 33 is detected and baseline statistics match the reference.

## Access or source problems

Use approved paired access or the local source files. Record unavailable live capabilities as not executed. If an attachment is not supported, paste the short synthetic text or reference its approved SharePoint copy. Do not use personal accounts to bypass controls.

# Lab 4.4 prompts and evidence

## Main prompt

Analyse the supplied synthetic fill dataset. Use samples 1-30 as baseline and 31-35 as monitoring. Estimate sigma with average successive baseline moving range divided by 1.128. Report n, mean, limits and provisional Cp/Cpk using 9.4 and 10.6 L. Detect monitoring signals and show calculations. Do not infer root cause from correlation. Separately summarise the defect Pareto.

## Verification prompt

Audit the draft against the named source files. List each claim, source location, uncertainty and correction. Recalculate key values independently. Return a revised draft and change log; do not approve the engineering deliverable.

## Expected output example

Illustrative teaching example, not a recorded AI run. Analysis reference: mean 10.01333, UCL 10.50856, LCL 9.51810 L. Sample 33 exceeds UCL. Research conclusion should be conditional on conductivity, pipe conditions and installation constraints; no universal winner is established.

## Retain your evidence

Keep the source versions, prompt, tool, output, corrections, independent checks and reviewer decision. Record any capability or permission test that was not executed. Use templates/prompt-log.csv. Do not publish real workplace data.